

EXP 9: Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

```
Untitled 1 - SciNotes
File Edit Format Options Window Execute ?
[Icons]
Untitled 1 - SciNotes
Untitled 1
1 clear;
2 clear;
3 theta_1=input("Enter the value of theta1=");
4 theta_2=input("Enter the value of theta2=");
5 medium_1=input("Enter the value of er1=");
6 tan_theta= tand(theta_1)/tand(theta_2);
7 medium_2= medium_1/tan_theta;
8 disp("THE VALUE OF ER2 IS", [medium_2]);
```

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser: C:\Users\kalev\Documents\

Scilab 2026.1.0 Console

```
Enter the value of theta1=30
Enter the value of theta2=60
Enter the value of er1=7
"THE VALUE OF ER2 IS"
21.000000
-->
```

Variable Browser

Name	Value	Type	Visibility	Memory
medium_1	7	Double	local	216 B
medium_2	21	Double	local	216 B
tan_theta	0.333	Double	local	216 B
theta_1	30	Double	local	216 B
theta_2	60	Double	local	216 B

Command History

```
-- 3
-- 4
-- 5
-- 6
-- 1
-- 2
-- 3
-- 5
-- 3
-- 6
-- 4
// -- 29/07/2026 11:46:54 -- //
-- 2
-- 8
-- 6
// -- 30/07/2026 13:28:28 -- //
// -- 03/08/2026 10:24:54 -- //
-- 30
-- 60
-- 7
```

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Exp 9:-

9) $\theta_1 = 30^\circ$

$\theta_2 = 60^\circ$

$\epsilon_{s1} = 7$

formula used is $\Rightarrow \frac{\tan \theta_1}{\epsilon_0} = \frac{\tan(\theta_2)}{\epsilon_{s2}}$

$\epsilon_{s2} = \frac{\epsilon_{s1} \tan \theta_1}{\tan \theta_2}$

$\tan(30^\circ) = 0.57735$

$\tan(60^\circ) = 1.73205$

$\frac{\tan(30^\circ)}{\epsilon_0} = \frac{\tan(60^\circ)}{\epsilon_{s2}}$

$\epsilon_{s2} = \frac{0.57735}{1.73205}$

$= 0.33333$

$\epsilon_{s2} = \frac{7}{0.33333}$

$= 21$

$\epsilon_{s2} = 21$

Case 1: In this case, the dielectric medium has a relative permittivity (ϵ_r) of 2, meaning the dielectric reduces the electric field by half compared to the free space. This is typical for materials like mica or certain types of plastics.

- **Free Space Electric Field (E1): 100 V/m**

- **Dielectric Electric Field (E2): 50 V/m**

Case 2: Here, the dielectric medium has a higher relative permittivity ($\epsilon_r=2.5$) compared to Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like glass or water.

- **Parameters:**

- o **E_{n1}=200 V/m (Electric field in free space).**

- o **E_{2n}=80 V/m (Electric field in the dielectric medium).**

Case 3: The dielectric medium here also has a relative permittivity of 2. The behavior of the material suggests it could be another low-polarization dielectric material, with a consistent reduction of the electric field by half.

- **Free Space Electric Field (E1_n): 150 V/m**

- **Dielectric Electric Field (E2_n): 75 V/m**

